

Community-level analysis

16S rRNA amplicon
sequences (n = 277)

m_main
diversity metric ~ group + age + sex + rainfall + diet
diversity + % fruit + (1|ID) + offset(log(reads))

m_full
diversity metric ~ *m_main* + temperature

Shotgun metagenomic
sequences (n = 71)

Temperature DAA

m_main
Taxon counts ~ group + age + sex + rainfall + %
fruit + (1|ID) + offset(log(reads))

m_full
Taxon counts ~ *m_main* + temperature

m_main
clr transformed pathway abundance ~ group +
age + sex + rainfall + % fruit + (1|ID)

m_full
clr transformed pathway abundance ~ *m_main* +
temperature

Neopterin * Temperature DAA

m_immune_main
Taxon counts ~ group + age + sex + rainfall +
% fruit + temperature + (1|ID) +
offset(log(reads))

m_immune_full
Taxon counts ~ *m_main* + temperature *
neopterin

m_immune_main
clr transformed abundance ~ group + age + sex
+ rainfall + % fruit + temperature + (1|ID)

m_immune_full
clr transformed abundance ~ *m_immune_main* +
temperature * neopterin

$\xrightarrow[\omega > 0.5]{m_full}$

$\xrightarrow[\omega > 0.5]{m_full}$